

Near-Field Transducer Design for Heat-Assisted Magnetic Recording Beyond 4 Tb/in²

Takuya Matsumoto, Niranjan Natekar, Wan Kuang, Barry Stipe

Western Digital Corporation, San Jose, CA 95119, USA
Takuya.Matsumoto@wdc.com

The near-field transducer (NFT) is the key optical component of the heat-assisted magnetic recording (HAMR) head that generates a nanoscale heating spot on the recording medium by exploiting surface plasmon resonance. Achieving areal density capability (ADC) beyond 4 Tb/in² requires reducing the cross-track thermal spot width below 20 nm, while maintaining sufficient NFT reliability under extreme thermal loading conditions. In this paper, we present NFT design considerations for next-generation HAMR heads based on electromagnetic (FDTD) simulations, thermal simulations, and micro-magnetic modeling. An optimized NFT with a reduced tip width and head-medium spacing provides a pathway toward >4 Tb/in² ADC.

Index Terms— Areal density capability, heat-assisted magnetic recording, near-field transducer, optical near-field, surface plasmon resonance,

I. INTRODUCTION

HEAT-assisted magnetic recording (HAMR) overcomes the magnetic recording trilemma — the conflict among thermal stability, writability, and signal-to-noise ratio that limits areal density capability — by transiently heating high-anisotropy FePt-based granular media near the Curie temperature during the write process. This temporarily reduces the coercivity to a level accessible by the write field and enables recording densities exceeding 2 Tb/in² [1], [2].

The near-field transducer (NFT) is the key optical component of the HAMR head that generates a nanoscale heating spot on the medium by exploiting surface plasmon resonance, confining optical energy far below the diffraction limit of light. In early HAMR recording demonstrations, a solid immersion lens [1] or a planar solid immersion mirror [3] was used, but the achievable optical spot size was limited by the diffraction limit of light. Near-field optical apertures could overcome the diffraction limit, but lacked optical efficiency. To generate an optical near-field with significantly improved optical efficiency, we proposed the use of a plasmonic antenna for recording devices in the early 2000s [4], and demonstrated recording feasibility using phase-change media [5] and magnetic bit-patterned media [6], [7]. We also proposed an integrated head design that couples an optical waveguide to the NFT via the evanescent field of the transverse magnetic (TM) waveguide mode [8].

Achieving areal recording densities beyond 4 Tb/in² imposes increasingly stringent requirements on the NFT. Assuming a bit aspect ratio between 2.0 and 2.3, the required track pitch is below 20 nm, and the cross-track thermal spot width on the recording medium must be below this value while maintaining sufficient reliability under extreme thermal loading conditions.

In this paper, we present NFT design considerations for next-generation HAMR heads based on electromagnetic (FDTD) simulations, thermal simulations, and micro-magnetic modeling.

II. HAMR HEAD STRUCTURE AND OPERATING PRINCIPLE

Fig. 1 shows schematic cross-sectional and perspective views of the NFT integrated with the magnetic write head. Light generated by a near-infrared laser diode is guided through a waveguide and coupled to the NFT placed near a main pole. The evanescent field of the TM waveguide mode excites surface plasmons on the NFT metal surface. The NFT flare concentrates surface-plasmon energy toward the tip, where charge accumulation generates a strongly confined optical near-field that locally heats the recording medium. The magnetic write head applies a write field to switch the magnetization of the recording medium while it is near its Curie temperature.

When charges accumulate at the tip, not only is a localized optical near-field generated around it, but substantial heat is also generated within the tip due to optical loss in the metal. Additionally, if any opaque material is transferred from the medium to the head surface near the tip, that material will also be heated and lead to further temperature rise. For efficient generation of surface plasmons, gold is used for its low optical loss, but it is prone to deformation at high temperatures. To avoid tip deformation and meet the required product lifetime, a robust material is used for the tip. A thermal shunt that contacts the NFT layer is also employed to improve reliability. The thermal shunt conducts heat from the tip region to a heat sink to reduce the NFT temperature.

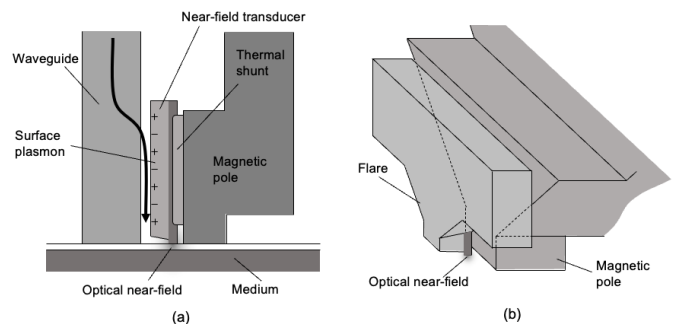


Fig. 1. Schematic of the integrated HAMR head: (a) cross-sectional view showing waveguide, NFT, thermal shunt, and magnetic pole, and (b) perspective view showing the flare structure

III. NFT DESIGN FOR >4 TB/IN²

The cross-track thermal spot size in the recording layer is mainly determined by the NFT tip width. To achieve areal densities beyond 4 Tb/in² with a track pitch below 20 nm, the NFT tip width must be significantly reduced. The required NFT tip width depends on the NFT geometry, material configuration, and surrounding structures. It also depends on head-medium spacing, the medium structure, and the medium's optical, thermal, and magnetic properties. When designing the NFT, these factors need to be carefully considered while maintaining sufficient reliability.

Fig. 2 shows the cross-track width of the thermal distribution in the recording layer as a function of NFT tip width for various head-medium spacings and NFT designs. The optical near-field is highly localized around the tip, and its width increases with distance from the tip. As shown by the dashed and dash-dotted lines, head-medium spacing must be reduced to achieve a narrow thermal spot width, especially when using a narrow tip. The achievable thermal spot width also depends on the NFT design: Design B, indicated by the solid line, achieves approximately 10% narrower cross-track thermal spot widths than Design A.

The achievable areal density also depends on the medium properties and recording conditions, and micro-magnetic modeling is required for its evaluation. In the talk, the achievable areal density will be discussed with micro-magnetic modeling results.

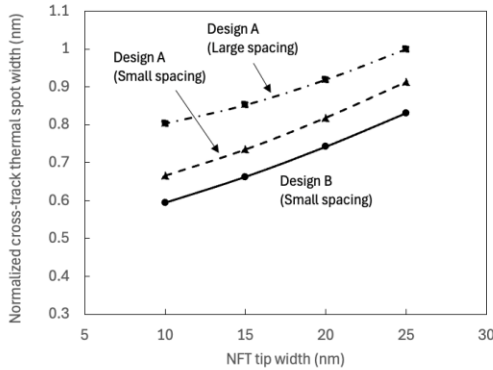


Fig. 2. Cross-track thermal spot width as a function of NFT tip width for Design A with large head-medium spacing, Design A with small head-medium spacing, and Design B with small head-medium spacing.

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